

Management

FORM FOR THE SUBMISSION OF ASSESSED GROUP COURSEWORK

THIS FORM MUST BE COMPLETED AND ATTACHED TO **ALL** ASSESSED WORK.

Please include all group members' student numbers below.

Student Number	2	7	4	4	1	0	1
Student Number	2	7	9	7	9	5	6
Student Number	2	7	7	5	1	8	9
Student Number	2	8	3	2	8	7	7
Student Number	2	7	2	8	0	6	0
Student Number	2	7	0	2	2	8	6

We agree that equal and fair contributions were made to the final report by all members of our group.

Please list brief details of how tasks were divided up among the group members in the box below. (Please refer to chapters of final report etc.)

- **Member 1:** Section 2 – Data Processing
- **Member 2:** Section 3.1 – DEA Model Formulation
- **Member 3:** Section 3.2 – Cross-Efficiency Analysis
- **Member 4:** Section 4 – Results and interpretation of Tables/Figures
- **Member 5:** Section 5 – Discussion and Section 7 – Limitations & Future Work
- **Member 6:** Section 6 – Conclusion and all Appendices (data sources, code summary, full code)

All members participated in editing, proofreading, and preparing the final submission.

By submitting this work online using my unique log-in and password, I confirm that I am submitting the work on behalf of my group and that it is all our own work.

We understand that all marks are provisional until ratified by the Faculty Examination Board.

Please remember to save **your work** either as a **Word document (.doc)** or as a **PDF (.pdf)** unless it is a presentation or recording where it should be saved in an appropriate format for the work with the filename [Unit Code_Group Number or Name] **e.g. MGRCM0000_Group 1.docx**.

Please start writing your coursework on the next page.

**Study of Formula 1 Constructors' Efficiency:
A DEA and Cross-Efficiency Approach**

Table of Contents

1. Introduction	1
2. Data and Variable Construction.....	1
2.1 Data Processing	1
2.2 Inputs and outputs.....	2
3. Methodology	3
3.1 DEA Model	3
3.2 Cross-efficiency Analysis.....	4
3.3 Constraints and Assumptions	4
4. Results	5
5. Discussion and Managerial Implications	9
6. Limitations and Future Research.....	10
7. Conclusion.....	10
References	11
Appendix	13

1. Introduction

Success in Formula 1 (F1) is not just about outright car speed; team principals and performance directors are judged on how efficiently they convert available performance into race results. The Data Envelopment Analysis (DEA) model has been extensively employed to analyse the budget-based efficiency of F1 constructors (Gutierrez & Lozano, 2014) and to analyse circuit designs in terms of speed, safety and environmental impact (Gutierrez & Lozano, 2020).

This report employs a Data Envelopment Analysis model with cross-efficiency based Decision Support System (DSS) to assess how F1 teams convert starting potential and mechanical reliability to points and regular finishing positions in the 2014-2023 hybrid era. Each constructor-year (team-season) is treated as a decision-making unit (DMU). A Constant Returns to Scale (CCR) DEA model (Charnes et al., 1978) is used to evaluate how efficiently two inputs i.e., average starting position and mechanical DNF (Did Not Finish) rate transform into two outputs: points per race and an average positional score. Cross-efficiency analysis (Doyle & Green, 1994) is utilised to peer-review the efficiency scores. In our DEA model, operational efficiency is measured by the self-efficiency score of each constructor-season. The findings provide operational benchmarks and recommendations for performance directors.

2. Data and Variable Construction

2.1 Data Processing

The data used is the Ergast F1 Database (Ergast Developer API, 2024), which contains race-by-race results, constructors and status codes from 1950 onwards. The specific data sources have been attached in the appendix.

Four tables from the dataset are merged:

- Results (finishing position, points, starting position)
- Races (events and season year)
- Constructors (teams)
- Status (DNF reasons)

The sample is restricted to the 2014-2023 hybrid era, yielding 102 constructor-year observations. Basic data cleaning checks confirm there are no missing or negative values for starting position, finishing position, points or status. Pit-lane starts (starting = 0) have been treated as 20th place to ensure that the model does not treat them as favourable starting positions.

2.2 Inputs and outputs

Each constructor-year DMU aggregates race-level information in a given year. The inputs and outputs were chosen to capture the chain from car potential to race outcomes using variables consistently recorded for all constructor-seasons.

Inputs (minimised)

1. Average starting position

- Position = 0 (pit-lane) is replaced with 20.
- Mean of this value across all entries in a season is used as the DMU's average starting position.
- Managerial meaning: the average starting position achieved by the car and qualifying operations. Lower values indicate better performance.

2. Mechanical DNF rate

- Retirement reasons that contain terms such as 'engine', 'power unit', 'gearbox', 'hydraulic' or similar mechanical descriptions are flagged as mechanical DNFs.
- Mechanical DNF rate = mechanical DNFs / total race entries
- Managerial meaning: reliability issues that the management can fix through engineering and quality processes.

Outputs (maximised)

1. Points per race

- Total constructor points divided by the number of races in that season, normalising for calendar length.

2. Average Positional score

- Finishing positions are mapped to a score such that P1 = 20, P2 = 19, and so on. DNFs and unclassified = 0. This is done so that it aligns with the DEA requirements of maximisation.

$$\text{Positional Score} = 21 - \text{Finishing position}$$

Constructor-years with zero points and zero positional score are dropped, so that they do not add any information to the frontier. With two inputs and outputs, the 102 DMUs provide adequate differentiation (Cooper et al., 2007).

Table 1: Summary of inputs and outputs

Inputs	Outputs
Average starting position (grid position)	Points per race
Mechanical DNF rate	Average positional score

3. Methodology

3.1 DEA Model

An input oriented CCR DEA model is used (Charnes et al., 1978), which assumes Constant Return to Scale (CRS). For each constructor-season, four variables are observed:

- X_{grid} - average starting position (input, lower is better)
- X_{fail} - mechanical DNF rate (input)
- X_{points} - points per race (output)
- X_{score} - average positional score (output)

For a given constructor-season, DEA chooses four non-negative weights to maximise its own efficiency:

- a and b on the outputs X_{points} and X_{score} ,
- c and d on the inputs X_{grid} and X_{fail} .

The DEA model chooses weights that maximises each constructor-season's weighted outputs, subject to its weighted inputs equalling one and no other constructor-season exceeding 100% efficiency under those weights. All constructor-year DMUs are analysed separately and all variables are normalised to the [0,1] range for numerical stability (Banker et al., 1984).

3.2 Cross-efficiency Analysis

Since classical DEA lets each constructor choose its own best weights, it can produce flattering but hard-to-compare results, so cross-efficiency analysis is applied (Doyle & Green, 1994). For each constructor-year i , its optimal weights (a_i, b_i, c_i, d_i) are reused to evaluate each constructor-year j .

$$E_{ij} = \frac{a_i \cdot X_{points,j} + b_i \cdot X_{score,j}}{c_i \cdot X_{grid,j} + d_i \cdot X_{fail,j}}$$

This results in an $N \times N$ matrix of cross-efficiency scores outlining how well each season performs under each other constructor's weight system. The column-wise average of this matrix shows an average cross-efficiency score ($\bar{E}_j$) for each constructor-season. Cross-efficiency analysis

$$\bar{E}_j = \frac{1}{N} \sum_{i=1}^N E_{ij}$$

can be interpreted as peer-reviewed efficiency score and has also been used in other sports benchmarking contexts (Gutierrez & Ruiz, 2013).

3.3 Constraints and Assumptions

The DEA model in this report assumes constant returns to scale i.e., proportional change in inputs are expected to require proportional change in outputs to maintain the efficiency. (Charnes et al., 1978). This is appropriate because constructors operate under similar technical regulations and cost caps. Instead of modelling uncertainty stochastically, each constructor - year DMU is analysed separately using observed values. All variables are normalised to [0,1]. Seasons with zero points and zero positional score are excluded from analysis since they do not provide any deterministic power. Pit-lane starts are recorded as 20th place to avoid overstating starting performance.

4. Results

Constructor Name	Year	Average Starting Position (X_{grid})	Mechanical DNF Rate (X_{fail})	Points Per Race (X_{points})	Average Positional Score (X_{score})
Ferrari	2014	7.947	0.026	11.368	12.579
	2015	6.395	0.026	22.526	14.974
	2016	5.69	0.048	18.952	13.619
	2017	3.95	0	26.1	15.525
	2018	3.31	0.048	27.19	16.071
	2019	4.262	0.024	24	15
	2020	10.265	0.029	7.706	10.294
	2021	7.182	0	14.659	13.886
	2022	4.318	0.091	23.591	14.318
	2023	5.773	0.045	16.5	13.136
McLaren	2014	8.632	0.053	9.526	12.053
	2015	15.842	0.158	1.421	6.526
	2016	12.571	0.143	3.619	8.333
	2017	13.85	0.225	1.5	7.15
	2018	14.405	0.119	2.952	8.238
	2019	9.81	0.071	6.905	10.31
	2020	8.559	0.029	11.882	12.382
	2021	8.273	0	12.455	13.205
	2022	9.727	0.045	6.818	10.364
	2023	8.795	0.045	12.091	11.545
Mercedes	2014	2.947	0.105	36.895	16.789
	2015	1.816	0	37	17.789
	2016	2.976	0.024	36.429	17.548
	2017	3.475	0.025	33.4	17.825
	2018	3.429	0.024	31.19	17.143
	2019	3.024	0.024	35.19	17.762
	2020	2.088	0.029	33.706	17.265
	2021	4.341	0	27.477	15.591
	2022	6.432	0.023	22.5	15.477
	2023	6.273	0.045	17	13.864
Red Bull	2014	7.026	0.105	21.316	14.289
	2015	8.895	0.105	9.842	12
	2016	5.286	0.024	22.286	15.286
	2017	7.05	0.2	18.4	12.35
	2018	7.619	0.119	19.952	13.357
	2019	6.762	0.048	19.857	14.738
	2020	5.206	0.059	18.765	13.294
	2021	5.455	0.023	26.295	15.159
	2022	4.114	0.023	32.909	16.545
	2023	6.227	0	35.909	17.432
Williams	2014	6.842	0.026	16.842	13.211
	2015	7.263	0.053	13.526	12.5
	2016	9.548	0.071	6.571	10.5
	2017	12.425	0.075	4.15	9.5
	2018	16.333	0.071	0.333	6.095
	2019	18.262	0.048	0.048	4.595
	2020	17.235	0.029	0	5.265
	2021	14.568	0.091	1.045	6.273
	2022	16.386	0.023	0.364	6.045
	2023	14.386	0.045	1.182	6.886

Table 2: Data of the top five constructors in 2014-2023
(All data rounded to three decimal places)

Table 2 summarises input and output data for the top five constructors (Mercedes, Ferrari, Red Bull, McLaren and Williams) from 2014 to 2023. Mercedes combine strong average starting

positions with low failure rates and high points per race. Ferrari and Red Bull show more variations, they both experience seasons with competitive starting positions but varying reliability and race execution. Williams and McLaren usually start further down the grid and score fewer points per race, although McLaren display slight improvement over time.

Figure 1 plots DEA self-efficiency scores for the same teams. Mercedes show sustained dominance, peaking at an efficiency of 1.0 in 2015 and remaining above 0.5 in most seasons, with a notable rebound in the 2020 season. Ferrari's efficiency peaks in 2017-2018 before declining sharply in 2020 and then partially recovering. While Red Bull's efficiency rises steadily, reaching a high in 2022 when car pace and race execution align. Williams continue to decline over the period, whereas McLaren improved from very low efficiency in 2015 to a more stable mid-field conversion by 2021-2022.

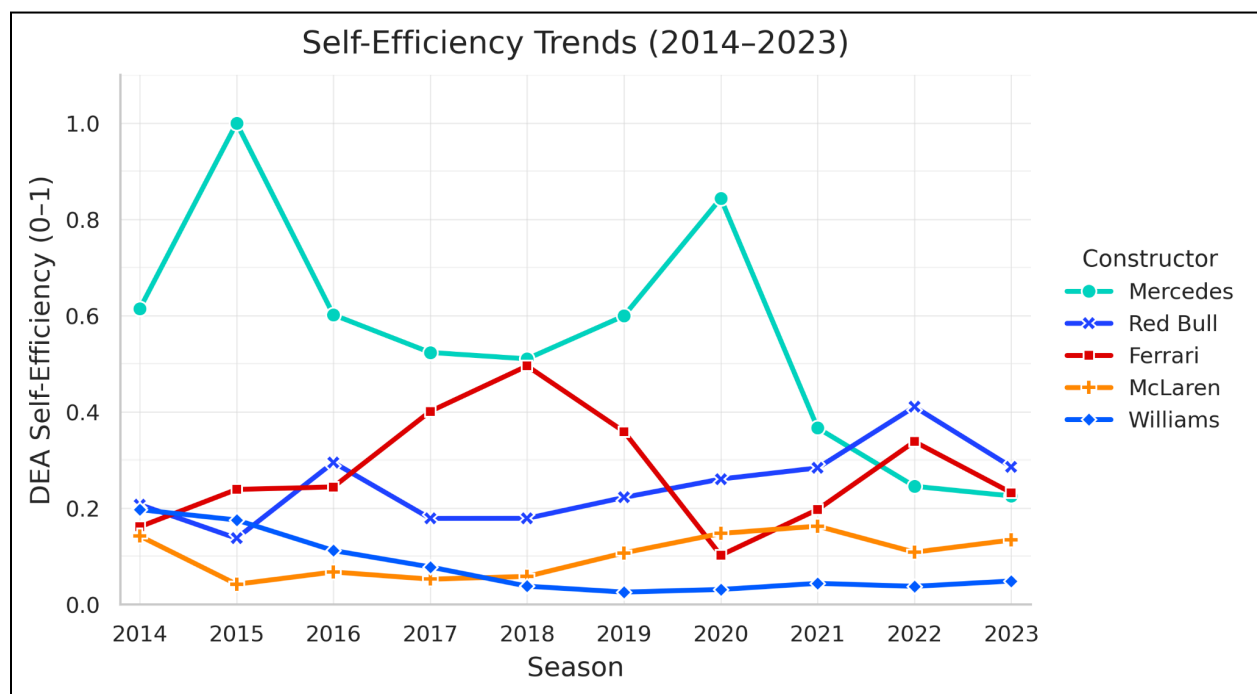


Figure 1: Self-efficiency trends for top five constructors during the period 2014-2023

DMU	Cross Efficiency Average	Efficiency	X_{grid}	X_{fail}	X_{points}	X_{score}
Mercedes - 2015	1	1	1.816	0	37	17.789
Mercedes - 2020	0.843	0.844	2.088	0.029	33.706	17.265
Mercedes - 2016	0.602	0.602	2.976	0.024	36.429	17.548
Mercedes - 2019	0.599	0.599	3.024	0.024	35.19	17.762
Mercedes - 2014	0.581	0.614	2.947	0.105	36.895	16.789
Mercedes - 2017	0.523	0.523	3.475	0.025	33.4	17.825
Mercedes - 2018	0.509	0.51	3.429	0.024	31.19	17.143
Ferrari - 2018	0.494	0.495	3.31	0.048	27.19	16.071
Red Bull - 2022	0.41	0.41	4.114	0.023	32.909	16.545
Ferrari - 2017	0.4	0.401	3.95	0	26.1	15.525

Table 3: DEA Self-efficiency and cross-efficiency scores for the top 10 DMUs (constructor-season)

Table 3 shows the top ten most efficient constructor-seasons sorted by the average cross-efficiency, along with their self-efficiency scores. As visible, Mercedes-2015 leads the frontier with both the self-efficiency and cross-efficiency scores equal to 1. The 2014, 2016, 2017, 2018 and 2019 Mercedes seasons also score highly, showing their consistent conversion of performance into results. Ferrari 2017 and 2018, and Red Bull 2022 are the only non-Mercedes seasons in the top ten.

Table 4 summarises results at the constructor level by averaging cross-efficiency, self-efficiency, and points per race across seasons. Mercedes rank first with the highest overall cross-efficiency score and average points per race, followed by Red Bull and Ferrari. Other teams trail on both the measures. This pattern is also consistent with budget-based DEA findings in F1, where a small group of teams also dominate efficiency rankings (Gutierrez & Lozano, 2014).

Constructor	Overall Cross Efficiency	Average Self Efficiency	Average points per race	Average Starting Position	Average Failure Rate	Average Positional Score	Overall Rank (Cross efficiency)	Overall Rank (Points)
Mercedes	0.549	0.553	31.079	3.68	0.03	16.705	1	1
Ferrari	0.276	0.277	19.259	5.909	0.034	13.94	2	3
Red Bull	0.245	0.246	22.553	6.364	0.071	14.445	3	2
Force India	0.111	0.112	7.638	10.103	0.049	11.011	4	5
Racing Point	0.105	0.105	7.915	11.266	0.095	10.504	5	4
McLaren	0.101	0.102	6.917	11.046	0.089	10.011	6	6
Alpine F1 Team	0.099	0.1	6.591	10.78	0.045	10.538	7	7
Aston Martin	0.086	0.087	6.03	12.333	0.03	10.068	8	8
Renault	0.083	0.083	4.804	11.881	0.111	8.867	9	9
Williams	0.078	0.078	4.406	13.325	0.053	8.087	10	10
AlphaTauri	0.074	0.074	3.835	12.32	0.072	8.721	11	11
Toro Rosso	0.065	0.066	2.729	12.787	0.134	8.211	12	12
Lotus F1	0.057	0.057	2.316	13.684	0.105	7.171	13	13
Alfa Romeo	0.054	0.054	1.382	14.381	0.042	7.6	14	15
Haas F1 Team	0.05	0.05	1.447	13.998	0.058	6.518	15	14
Sauber	0.043	0.043	0.905	15.657	0.085	6.5	16	16
Marussia	0.023	0.023	0.105	18.677	0.097	4.258	17	17
Manor Marussia	0.023	0.023	0.024	18.571	0.012	4.194	17	18
Caterham	0.017	0.017	0	19.441	0.206	3.265	19	19

Table 4: Constructor-level comparison of overall cross-efficiency, self-efficiency and points

As can be seen from the data, across all constructor-seasons, DEA self-efficiency and average cross-efficiency are highly aligned. Figure 2 shows a strong positive relationship between the both of them, with most points lying close to the 45-degree line. This affirms that the most efficient seasons under their own weights also appear efficient under peers' weight systems, indicating that the efficient frontier is robust and not driven by extreme, DMU-specific weight choices, which supports the use of cross-efficiency as a stability check rather than a separate efficiency analysis.

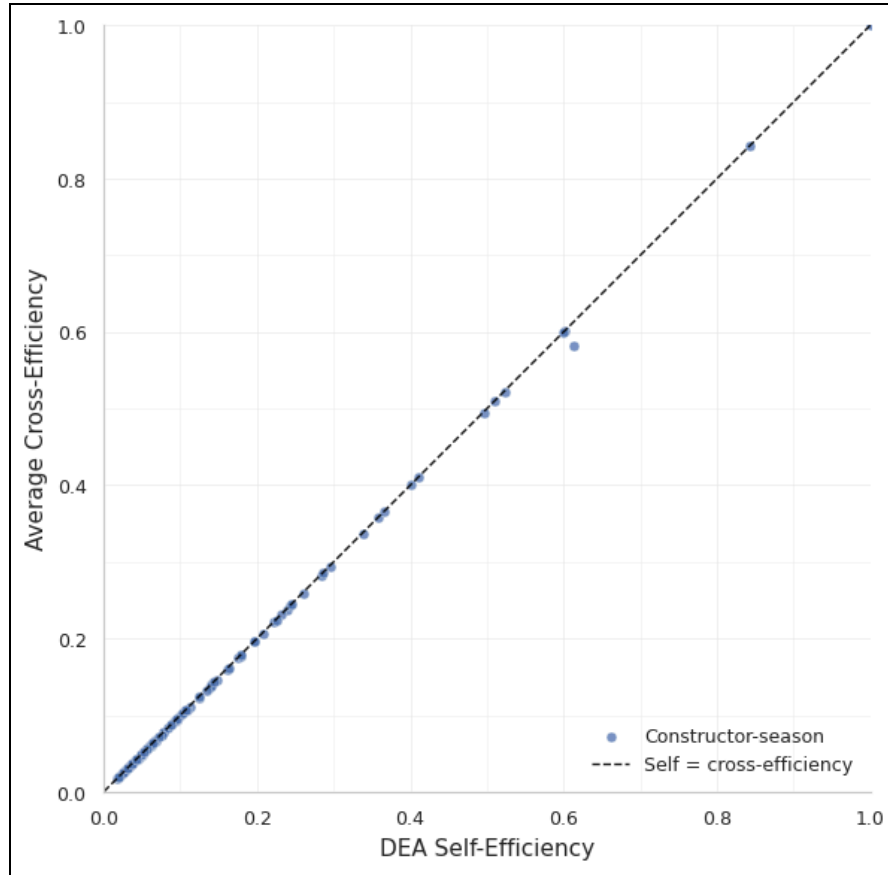


Figure 2: Relationship between DEA self-efficiency and cross-efficiency

5. Discussion and Managerial Implications

Beyond the top performing constructor-seasons, the DEA results mainly serve as a benchmarking tool for the remainder of the grid. For mid-field and lower-ranked constructors, the combination of average grid position, failure rates and efficiency scores shows whether they are extracting value from the car they have. Constructors that start at the back but achieve high efficiency can be seen as value creators, while those that start near the front yet post low efficiency scores are potential wasters of underlying pace.

The weight structure also carries clear managerial implications. The model assigns significant importance to both points per race and positional score, but the mechanical DNF rate is penalised more than starting position. This indicates that, once a constructor becomes roughly competitive

on one-lap pace, reducing avoidable mechanical failures may deliver more efficiency gains than marginal qualifying improvements.

6. Limitations and Future Research

First, the model only uses sporting variables (starting position, mechanical DNF rate, points per race and positional score) and excludes financial or cost-cap data since such data for each constructor-season are not openly available, so it measures operational efficiency rather than the full financial efficiency. Second, the DEA model is static: each constructor-season is evaluated independently, and carry-over between seasons is not captured; a dynamic DEA model could capture how constructors move over time. Finally, reliable data exist for starting position, mechanical DNFs, points and finishing positions, whereas factors such as pit stops, tyre strategy, weather and safety cars are noisy or inconsistently recorded and therefore only indirectly reflected in the outputs.

7. Conclusion

In conclusion, this study evaluates how efficiently F1 constructors convert starting positions and reliability into race results in the 2014-2023 hybrid era. While a small number of constructor-seasons define the efficiency frontier, the main value of the framework is in benchmarking the rest of the grid and identifying where the gaps arise from pace, reliability or execution, and whether a team behaves as a value creator or a potential waster.

For the team principals and performance directors, the framework turns publicly available race data into a transparent set of operational metrics that can support performance reviews, process improvements and communication with boards, sponsors and drivers, even without detailed budget data.

References

Banker, R.D., Charnes, A. and Cooper, W.W. (1984) ‘Some models for estimating technical and scale inefficiencies in data envelopment analysis’, *Management Science*, 30(9), pp. 1078–1092.

Available at: <https://doi.org/10.1287/mnsc.30.9.1078> (Accessed: 1 December 2025).

Charnes, A., Cooper, W.W. and Rhodes, E. (1978) ‘Measuring the efficiency of decision making units’, *European Journal of Operational Research*, 2(6), pp. 429–444. Available at:

<https://www.sciencedirect.com/science/article/pii/0377221778901388> (Accessed: 1 December 2025).

Cooper, W.W., Seiford, L.M. and Tone, K. (2007) *Data Envelopment Analysis: A comprehensive text with models, applications, references and DEA-Solver software*. 2nd edn. New York:

Springer. Available at: <https://doi.org/10.1007/978-0-387-45283-8> (Accessed: 1 December 2025).

Doyle, J.R. and Green, R.H. (1994) ‘Efficiency and cross-efficiency in DEA: derivations, meanings and uses’, *Journal of the Operational Research Society*, 45(5), pp. 567–578. Available

at: <https://doi.org/10.1057/jors.1994.84> (Accessed: 1 December 2025).

Ergast Developer API (2024) *Ergast F1 API documentation*. Available at:

<http://ergast.com/api/f1> (Accessed: 1 December 2025).

Gutiérrez, E. and Lozano, S. (2014) ‘A DEA approach to performance-based budgeting of Formula One constructors’, *Journal of Sports Economics*, 15(2), pp. 180–200. Available at:

<https://doi.org/10.1177/1527002512447629> (Accessed: 1 December 2025).

Gutiérrez, E. and Lozano, S. (2020) ‘Benchmarking Formula One auto racing circuits: a two stage DEA approach’, *Operational Research*, 20(4), pp. 2059–2083. Available at:

<https://doi.org/10.1007/s12351-018-0416-z> (Accessed: 1 December 2025).

Gutiérrez, Ó. and Ruiz, J.L. (2013) ‘Data Envelopment Analysis and cross-efficiency evaluation in the management of sports teams: The assessment of game performance of players in the Spanish handball league’, *Journal of Sport Management*, 27(3), pp. 217–229. Available at:

<https://journals.humankinetics.com/view/journals/jsm/27/3/article-p217.xml> (Accessed: 1 December 2025).

Appendix

Appendix 1

Summary of all the model notations

Symbol	Meaning
N	Number of DMUs (constructor–seasons) in the sample
j	Index of the DMU being evaluated, (j = 1,...,N)
i	Index of the evaluating DMU in the cross-efficiency matrix, (i = 1,...,N)
m	Number of inputs in the DEA model (here m = 2)
s	Number of outputs in the DEA model (here s = 2)
x_{rj}	Value of input r for DMU j (e.g. average grid position, failure rate)
y_{qj}	Value of output q for DMU j (e.g. points per race, positional score)
v_r	Weight assigned to input r (to be optimised by DEA)
u_q	Weight assigned to output q (to be optimised by DEA)
E_j	Self-efficiency score of DMU j under its own optimal weights
E_{ij}	Cross-efficiency of DMU j evaluated using DMU i's optimal weights
$\bar{E}_j$	Average cross-efficiency of DMU j across all evaluators (i=1,...,N)

Appendix 2

Self-efficiency

For each constructor-season j, the constant-returns-to-scale DEA model solves:

$$\max_{u,v} E_j = \frac{\sum_{q=1}^s u_q y_{qj}}{\sum_{r=1}^m v_r x_{rj}}$$

subject to

$$\frac{\sum_{q=1}^s u_q y_{qk}}{\sum_{r=1}^m v_r x_{rk}} \leq 1 \quad \forall k = 1, \dots, N$$

$$\sum_{r=1}^m v_r x_{rj} = 1, \quad u_q \geq \varepsilon, \quad v_r \geq \varepsilon$$

where ε is a small positive constant ensuring strictly positive weights.

In the implementation:

- Inputs: $x1j_{x_{\{1j\}}}$ = normalised average grid position, $x2j_{x_{\{2j\}}}$ = normalised mechanical failure rate.
- Outputs: $y1j_{y_{\{1j\}}}$ = normalised points per race, $y2j_{y_{\{2j\}}}$ = normalised average positional score.

Cross-efficiency

Given optimal weights ($u^{(i)}, v^{(i)}$) obtained when unit i maximises its own self-efficiency, the cross-efficiency of DMU j evaluated with DMU i 's weights is:

$$E_{ij} = \frac{\sum_{q=1}^s u_q^{(i)} y_{qj}}{\sum_{r=1}^m v_r^{(i)} x_{rj}}$$

The average cross-efficiency for DMU j is then:

$$\bar{E}_j = \frac{1}{N} \sum_{i=1}^N E_{ij}$$

This matrix E_{ij} allows us to compare how each constructor–season performs under its own preferred weights and under the weight systems chosen by all other seasons.

Appendix 3

Data Sources: All race-level performance data were taken from the Formula 1 World Championship (1950-2023) Race Data dataset on Kaggle (Ergast F1 database)

Link: <https://www.kaggle.com/datasets/jtrotman/formula-1-race-data>

- races.csv – calendar and season information (year, raceId, race name).
- results.csv – driver-level race results (grid position, finishing position, points).
- constructors.csv – constructor identifiers and team names.
- status.csv – textual description of race retirement status.

Appendix 4

Generative AI Use Disclosure

In preparing this assignment, we used the following ChatGPT and Gemini in the following ways:

- Used for brainstorming ideas, and ways to structure the DEA model.
- Learning and checking understanding of DEA and cross-efficiency concepts.
- Verifying our code and cleaning it for understanding and adding comments.
- Summarising Literature related to the paper and verifying references.
- Grammar check and structure verification.

Whenever AI tools suggested explanations, code fragments or analytical ideas, we treated them as draft input only.

Appendix 5

Link to the Code:  **Modelling Analytics Formula 1.ipynb**

The model is implemented in Python using PuLP optimisation library.